

The Sharp Coefficient Radius of the Deterministic Numerator

Bin Wang

July 2026

Abstract

Let $q_* = (r_H \lambda)^{-1}$. We prove the sharp all-order law $a_n/q_*^n \rightarrow 1$ for the Hardy-scaled deterministic numerator. Hence the logarithmic coefficient root rate is exactly q_* , the convergence radius is exactly q_*^{-1} , no smaller geometric base is possible, and the absolute logarithmic series diverges at the critical radius. These are deterministic target statements, not moving-cloud asymptotics.

1 Odd and even limits

For odd $n \geq 3$, the exact factor formula is [2, 3]

$$\frac{a_n}{q_*^n} = \frac{1}{1 + \lambda^{-n}} \rightarrow 1.$$

For even $n = 2m$, RH-263 gives

$$\frac{a_{2m}}{q_*^{2m}} = 2\lambda^{2m} \operatorname{tr}(T^m) + \frac{1 - 2\lambda^{-m}}{1 - \lambda^{-2m}}.$$

If $m = 3k + j$, the RH-13 trace-ideal estimate [1] gives

$$|\lambda^{2m} \operatorname{tr}(T^m)| \leq \nu_j \lambda^{2j} (q_3 \lambda^6)^k, \quad q_3 \lambda^6 < 0.801254.$$

Thus the trace term tends to zero, while the endpoint fraction tends to one.

Theorem 1.1 (Sharp coefficient law). *For the complete sequence $n \geq 2$,*

$$\boxed{\lim_{n \rightarrow \infty} \frac{a_n}{q_*^n} = 1.}$$

2 Radius and negative consequences

Corollary 2.1 (Exact radius). *The power series $-\sum_{n \geq 2} a_n z^n / n$ has exact radius*

$$\rho_* = q_*^{-1} = r_H \lambda = 1.4267874838640739 \dots$$

Proof. Since $n^{1/n} \rightarrow 1$, the coefficient root rate of a_n/n is q_* . The Cauchy–Hadamard formula gives the radius. $\square$

Corollary 2.2 (Sharp scoped negative result). *There are no constants $C < \infty$ and $q < q_*$ such that $|a_n| \leq Cq^n$ for all n . Moreover*

$$\sum_{n \geq 2} \frac{|a_n| \rho_*^n}{n} = \infty.$$

Proof. The first claim contradicts the exact root rate. For the second, $|a_n|\rho_*^n \rightarrow 1$, so the terms are asymptotic to the harmonic series. $\square$

The order-2–28 ratios approach one numerically, but no finite fit is used in the proof. The sharp rate belongs only to the deterministic target. It does not provide a cloud coefficient bridge, legal selector, or uniform quotient tail. Gates A–E and all Hilbert–Polya/zeta/RH claims remain false/open.

References

- [1] Bin Wang. Validated reduced-sector spectral gaps. *Preprint*, 2026. RH-13 preprint.
- [2] Bin Wang. Parity-extracted bulk scattering. *Preprint*, 2026. RH-15 preprint.
- [3] Bin Wang. A certified unified envelope for deterministic numerator traces. *Preprint*, 2026. RH-267 preprint.